Autodeformalization of CMOS Sensor and Cramér-Rao Bounds

1 Introduction

This document provides a human-readable translation of the formally verified Lean 4 theorems and definitions regarding the modeling of CMOS sensor signals and the corresponding Cramér-Rao Lower Bounds (CRLB) for unbiased estimators.

2 Cramér-Rao Lower Bound Framework

The underlying mathematical framework relies on measuring the theoretical limit of variance for any unbiased estimator.

Definition 1 (Parameter Space)**.** *Let ι be a finite index set. The parameter space is defined as $\Theta = \mathbb{R}^{|\iota|}$, where each parameter vector is $\theta \in \Theta$.*

Definition 2 (Fisher Information Matrix)**.** *Given raw sensor data $x \in Raw$, an expected forward signal model $g(\theta, x)$, and a noise variance $\sigma(x)^2$, the Fisher Information Matrix $F(x) \in \mathbb{R}^{|\iota|\times|\iota|}$ for a single pixel x using the canonical basis is formulated as:*

$$F(x)_{ij} = \frac{\frac{\partial g(\theta,x)}{\partial \theta_i} \frac{\partial g(\theta,x)}{\partial \theta_j}}{g(\theta, x) + \sigma(x)^2} \tag{1}$$

Definition 3 (Image Fisher Information)**.** *For an entire array of pixels, denoted as the Field of View (FOV), the total Fisher Information Matrix is the sum of the individual pixel information matrices:*

$$F_{image}(\theta) = \sum_{x \in FOV} F(x)_{ij} \tag{2}$$

Definition 4 (Unbiased Estimator)**.** *An estimator $d : (Raw \to \mathbb{R}) \to \Theta$ is unbiased under the measure μ if its expected value equates to the true parameter θ:*

$$\mathbb{E}_\mu[d] = \int m \, d(m) \, d\mu = \theta \tag{3}$$

Definition 5 (Covariance Matrix)**.** *The covariance matrix $C \in \mathbb{R}^{|\iota|\times|\iota|}$ of the estimator d around the true parameter θ is the expected value of the outer product of the error vector:*

$$C = \mathbb{E}_\mu \left[(d(m) - \theta)(d(m) - \theta)^T \right] \tag{4}$$

Theorem 1 (Cramér-Rao Inequality)**.** *For any unbiased estimator d measuring parameters θ, with covariance matrix C and an invertible total Fisher Information Matrix F, the matrix difference $C - F^{-1}$ is positive semi-definite (i.e., $C \succeq F^{-1}$).*

3 Extended CMOS Sensor Model

We translate the abstract Fisher Information concepts into a concrete physical system representing a generalized CMOS sensor, extending the work of Gerwe and Idell (2003).

Definition 6 (Parameter Set)**.** *The target estimation parameter set includes:*

- α*: Orientation angle*
- x_c, y_c, z_c*: Translations in 3D space*
- N*: Total expected signal photoelectrons*
- b*: Expected background photoelectrons per pixel*

Thus, $\theta = (\alpha, x_c, y_c, z_c, N, b)^T$*.*

Definition 7 (Forward Model)**.** *Let* $s_m(\alpha, x_c, y_c, z_c)$ *be the normalized signal at pixel* m*. The expected total photoelectron count (the forward model)* $g_m(\theta)$ *is:*

$$g_m(\theta) = N s_m(\alpha, x_c, y_c, z_c) + b \tag{5}$$

4 Asymptotic Standard Error Regimes

To evaluate sensor performance across varied integration times T, the standard error scalar formulation is defined as:

$$\mathrm{SE}(T) = \sqrt{\frac{\Phi T s_m + \beta T + \sigma^2}{\Phi^2 T^2 s_m^2}} \tag{6}$$

where Φ is the signal flux, β is the background flux, s_m is the signal profile, and σ^2 is the read noise variance.

Lemma 1 (Regime 1: Photon-Noise Limited)**.** *Assuming* $\Phi > 0$*,* $s_m > 0$*, and* $\beta \geq 0$*, as the integration time approaches infinity (*$T \to \infty$*), the standard error is governed by photon counting statistics and scales as:*

$$SE(T) = \Theta\left(\frac{1}{\sqrt{T}}\right) \tag{7}$$

Lemma 2 (Regime 2: Read-Noise Limited)**.** *Assuming* $\Phi > 0$*,* $s_m > 0$*, and* $\sigma^2 > 0$*, for exceptionally short integration times as* $T \to 0$*, the readout noise* σ^2 *dominates the variance. The standard error scales inversely with* T*:*

$$SE(T) = \Theta\left(\frac{1}{T}\right) \tag{8}$$

5 References

- Gerwe, D. R., & Idell, P. S. (2003). Cramér-Rao analysis of orientation estimation: viewing geometry influences on the information conveyed by target features. *Journal of the Optical Society of America A*, 20(5), 797-816. [Validates the parameter definitions and foundational Cramér-Rao estimation context for imaging sensors].